

Brian Gakuru

Dedan Kimathi, Nyeri

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GitHub: github.com/kimanirian/Projects

Professional Summary

Analytical and results-driven mathematician with expertise in **data analytics, machine learning, and applied mathematical modelling**. Proficient in **Python, SQL, and deep learning frameworks**, with proven ability to build predictive models, extract insights, and translate findings into actionable strategies. Experienced in teaching, research, and project execution, with strong leadership and teamwork skills.

Education

- **Postgraduate Diploma in TVE** (*Ongoing, 2025 – Present*)
Dedan Kimathi University of Technology, Nyeri
 - **Bachelor of Science in Mathematics & Modelling Processes** (*2021 – 2024*)
Dedan Kimathi University of Technology, Nyeri
Award: Second Class Honours
 - **Short Course: Generative AI** (*Ongoing*)
Open University of Kenya (OUK)
 - **Manage Successful Field Research(Short Course)-** The World Bank Group Institute for Economic Development(*Ongoing*)
 - **High School Education** (*2017 – 2020*)
Jomo Kenyatta Boys High School & St. Xaviers High School
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Technical Skills

- **Programming & Tools:** Python(Advanced & OOP), SQL(PostGres & MySQL), Git, Jupyter, Excel, C(Foundation background), Data Structures & Algorithms(Python and C)
- **Data Analysis & Visualization:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Streamlit(Dashboards and Data Apps), PowerBI

- **Machine Learning:** Scikit-learn, TensorFlow, PyTorch, Regression Analysis, Clustering, Neural Networks (CNN, RNN, LSTMs)
 - **Mathematics & Statistics:** Algebra, Calculus, Linear Algebra(I & II), Probability & Statistics(I & II), Stochastic Processes, Differential Equations, Applied Modelling, Discrete Mathematics, Computational Mathematics
 - **Docker** – containerization and orchestration of projects
 - **Data pipeline orchestration-** Data Version control(DVC)
 - **Generative AI(Langchain & LangGraph)** – used for agentic AI and chatbot creation using LLMs
 - **Github-** for version control
 - **FastAPI-** for orchestration of endpoints for ML pipelines and models and creation of CRUD architecture.
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Professional Experience

Graduate Assistant Trainee (Volunteer)

Dedan Kimathi University of Technology – May 2025 – Present

- Assisting in teaching and research in applied mathematics and data science.
- Supporting undergraduate students in analytics and programming projects.

Data Analyst Intern

Data Science & Artificial Intelligence Lab (DSAIL), Dedan Kimathi University – Oct 2023 – Dec 2023

- Conducted **data analysis using Python** to generate insights for strategic decision-making.
- Built **predictive models** for customer behavior and trend forecasting.
- Collaborated with academic and research teams to implement analytics solutions.

Hackathons & Competitions

- Participated in Umoja Hackathon and Zindi challenges focused on data science and AI, strengthening problem-solving and teamwork skills.

Invigilator

Worked as an invigilator to Summative examinations at Nyeri National Polytechnic.

Projects

- **Sales Forecasting Model** – Built a time-series forecasting model for sales trends, enhancing resource allocation.
- **Customer Segmentation** – Applied clustering algorithms to identify market segments for targeted strategies.
- **Robbery Rate Prediction** – Developed ARIMA-based models for time-series forecasting of monthly crime data.
- **Multiple Regression (Boston Housing)** – Trained and tuned regression models to predict housing prices.
- **English–German Translation Model** – Implemented LSTM sequence-to-sequence model for machine translation.
- **Patient Admission Classification** – Designed ML model to predict patient admission/readmission, supporting medical planning.
- **Household Power Consumption Forecasting** – Created time-series models for energy demand prediction.
- **Streamlit Dashboards**- Designed dashboards that can be used to give real life insights and reports

Full project details available on [GitHub](#).

Leadership & Extracurricular Activities

- **Chairperson, Mathematics Club Committee (2023)**
 - **Instruments & Assets Coordinator, DekutCU (2023 – 2024)**
 - **Member, Constitution Review Committee, DekutCU (2024)**
 - **Chairperson, Future Tycoons Savings Help Group (2023 – Present)**
 - **Community Service, mentorship session for highschoolers, charity activities**
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References

Available upon request.